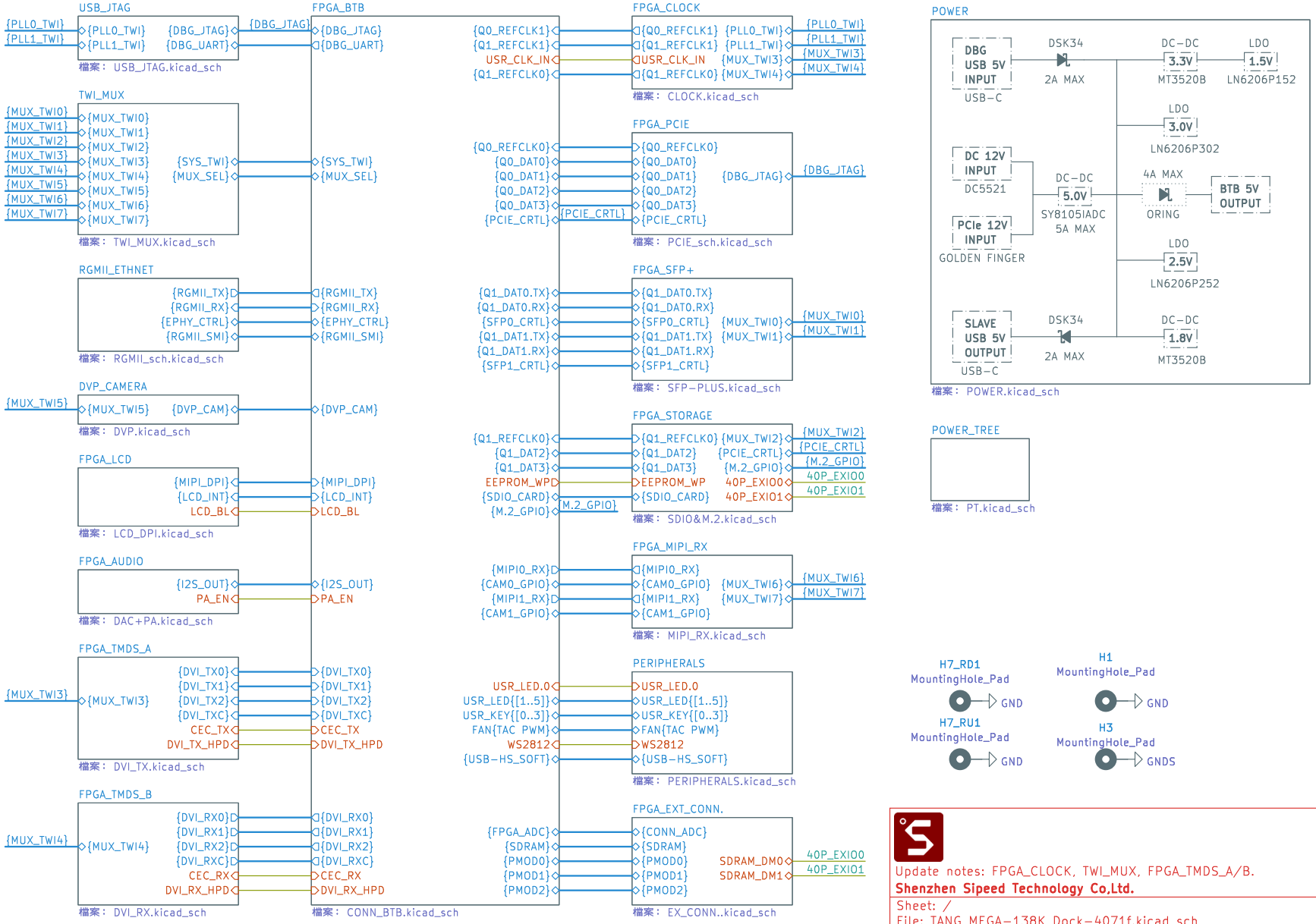
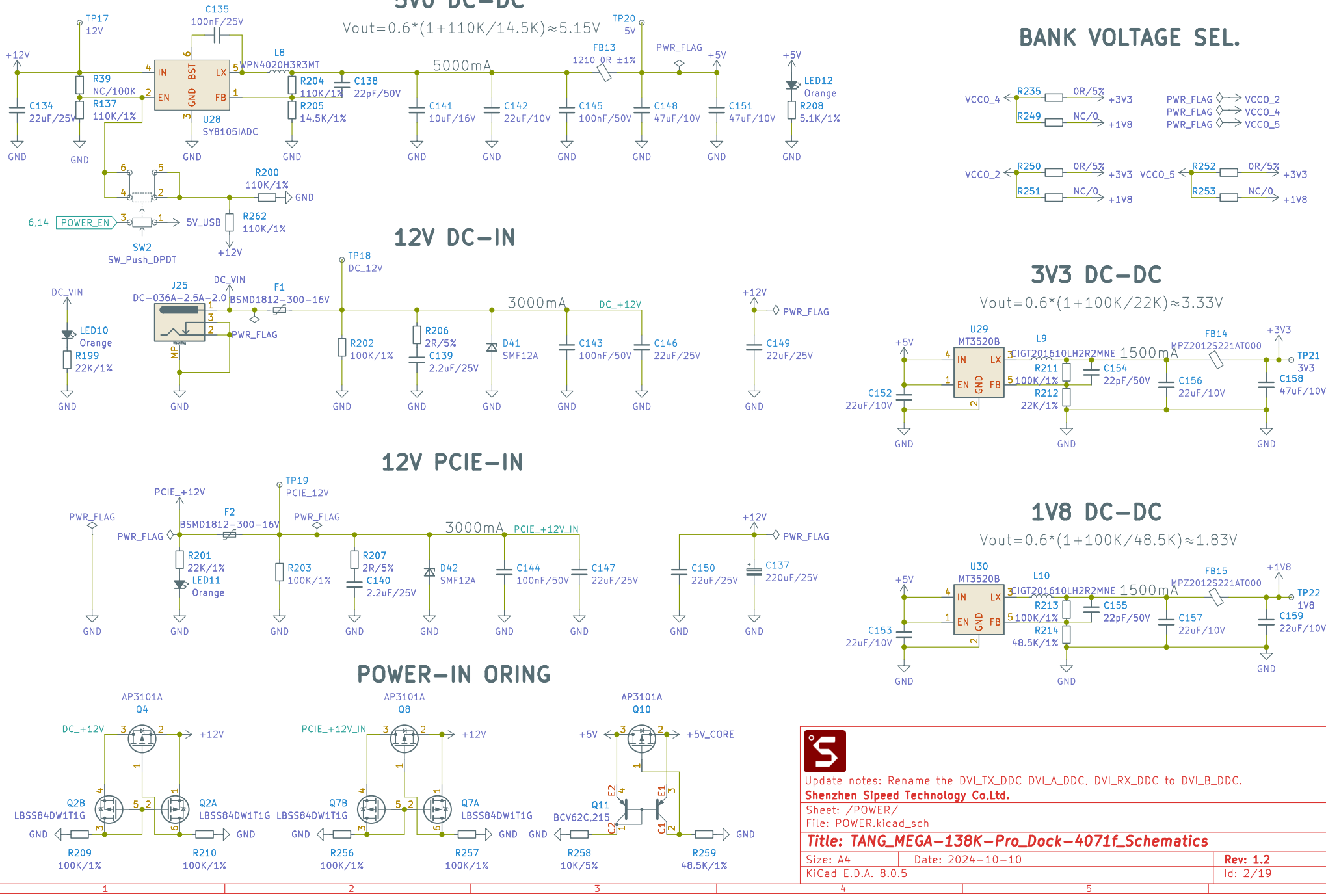

USB_JTAG
{PLL0_TWI}
{PLL1_TWI}
{DBG_JTAG}
{DBG_UART}
檔案： USB_JTAG.kicad_sch
FPGA_BTB
{DBG_JTAG}
{DBG_UART}
TWI_MUX
{MUX_TWI0}
{MUX_TWI1}
{MUX_TWI2}
{MUX_TWI3}
{MUX_TWI4}
{MUX_TWI5}
{MUX_TWI6}
{MUX_TWI7}
{SYS_TWI}
{MUX_SEL}
檔案： TWI_MUX.kicad_sch
RGMII_ETHNET
{RGMII_TX}
{RGMII_RX}
{EPHY_CTRL}
{RGMII_SMI}
檔案： RGMII_sch.kicad_sch
DVP_CAMERA
{MUX_TWI5}
{DVP_CAM}
檔案： DVP.kicad_sch
FPGA_LCD
{MIPI_DPI}
{LCD_INT}
LCD_BL
檔案： LCD_DPI.kicad_sch
FPGA_AUDIO
{I2S_OUT}
PA_EN
檔案： DAC+PA.kicad_sch
FPGA_TMDS_A
{MUX_TWI3}
{DVI_TX0}
{DVI_TX1}
{DVI_TX2}
{DVI_TXC}
CEC_TX
DVI_TX_HPD
檔案： DVI_TX.kicad_sch
FPGA_TMDS_B
{MUX_TWI4}
{DVI_RX0}
{DVI_RX1}
{DVI_RX2}
{DVI_RXC}
CEC_RX
DVI_RX_HPD
檔案： DVI_RX.kicad_sch
檔案： CONN_BTB.kicad_sch
FPGA_CLOCK
{Q0_REFCLK1}
{Q1_REFCLK1}
USR_CLK_IN
{Q1_REFCLK0}
{PLL0_TWI}
{PLL1_TWI}
{MUX_TWI3}
{MUX_TWI4}
檔案： CLOCK.kicad_sch
FPGA_PCIE
{Q0_DAT0}
{Q0_DAT1}
{Q0_DAT2}
{Q0_DAT3}
{PCIE_CRTL}
{DBG_JTAG}
檔案： PCIE_sch.kicad_sch
FPGA_SFP+
{Q1_DAT0.TX}
{Q1_DAT0.RX}
{SFP0_CRTL}
{Q1_DAT1.TX}
{Q1_DAT1.RX}
{SFP1_CRTL}
{MUX_TWI0}
{MUX_TWI1}
檔案： SFP-PLUS.kicad_sch
FPGA_STORAGE
{Q1_DAT2}
{Q1_DAT3}
EEPROM_WP
{SDIO_CARD}
{M.2_GPIO}
{MUX_TWI2}
40P_EXIO0
40P_EXIO1
檔案： SDIO&M.2.kicad_sch
FPGA_MIPI_RX
{MIPI0_RX}
{CAM0_GPIO}
{MIPI1_RX}
{CAM1_GPIO}
{MUX_TWI6}
{MUX_TWI7}
檔案： MIPI_RX.kicad_sch
PERIPHERALS
USR_LED.0
USR_LED{[1..5]}
USR_KEY{[0..3]}
FAN{TAC PWM}
WS2812
{USB-HS_SOFT}
檔案： PERIPHERALS.kicad_sch
FPGA_EXT_CONN.
{FPGA_ADC}
{SDRAM}
{PMOD0}
{PMOD1}
{PMOD2}
{CONN_ADC}
SDRAM_DM0
SDRAM_DM1
檔案： EX_CONN..kicad_sch
POWER
DBG USB 5V INPUT
USB-C
DSK34
2A MAX
DC-DC
3.3V
MT3520B
LDO
1.5V
LN6206P152
LDO
3.0V
LN6206P302
DC 12V INPUT
DC5521
PCIe 12V INPUT
GOLDEN FINGER
DC-DC
5.0V
SY8105IADC
5A MAX
4A MAX
ORING
BTB 5V OUTPUT
LDO
2.5V
LN6206P252
SLAVE USB 5V OUTPUT
USB-C
DSK34
2A MAX
DC-DC
1.8V
MT3520B
檔案： POWER.kicad_sch
POWER_TREE
檔案： PT.kicad_sch
H7_RD1
MountingHole_Pad
GND
H1
MountingHole_Pad
GND
H7_RU1
MountingHole_Pad
GND
H3
MountingHole_Pad
GNDS
Update notes: FPGA_CLOCK, TWI_MUX, FPGA_TMDS_A/B.
Shenzhen Sipeed Technology Co,Ltd.
Sheet: /
File: TANG_MEGA-138K_Dock-4071f.kicad_sch
Title: TANG_MEGA-138K-Pro_Dock-4071f_Schematics
Size: A4
Date: 2024-10-10
Rev: 1.2
KiCad E.D.A. 8.0.5
Id: 1/19



Update notes: Rename the DVI_TX_DDC DVI_A_DDC, DVI_RX_DDC to DVI_B_DDC.

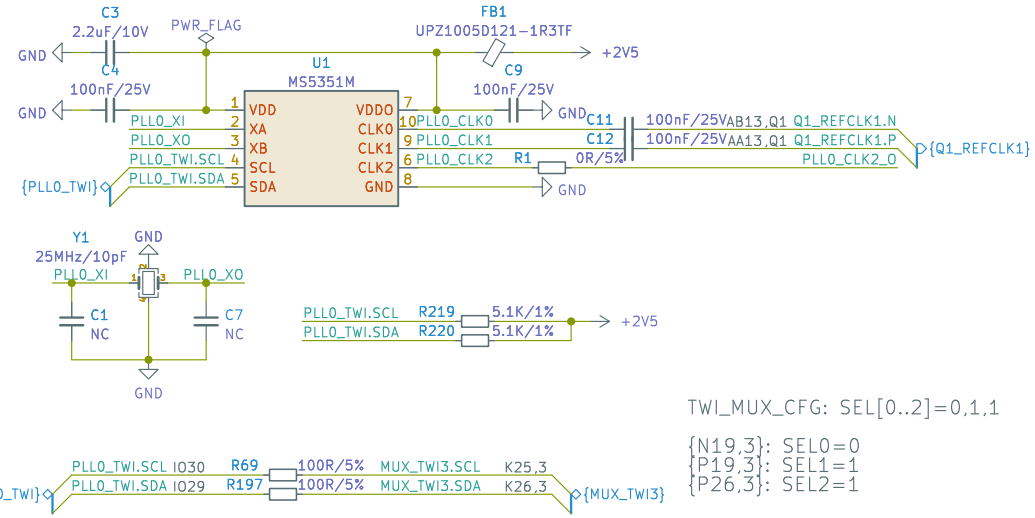
Shenzhen Sipeed Technology Co,Ltd.

Sheet: /POWER/		
File: POWER.kicad_sch		
Title: TANG_MEGA-138K-Pro_Dock-4071f_Schematics		
Size: A4	Date: 2024-10-10	**Rev: 1.2**
KiCad E.D.A. 8.0.5		Id: 2/19

CLOCK

EXT. PLL

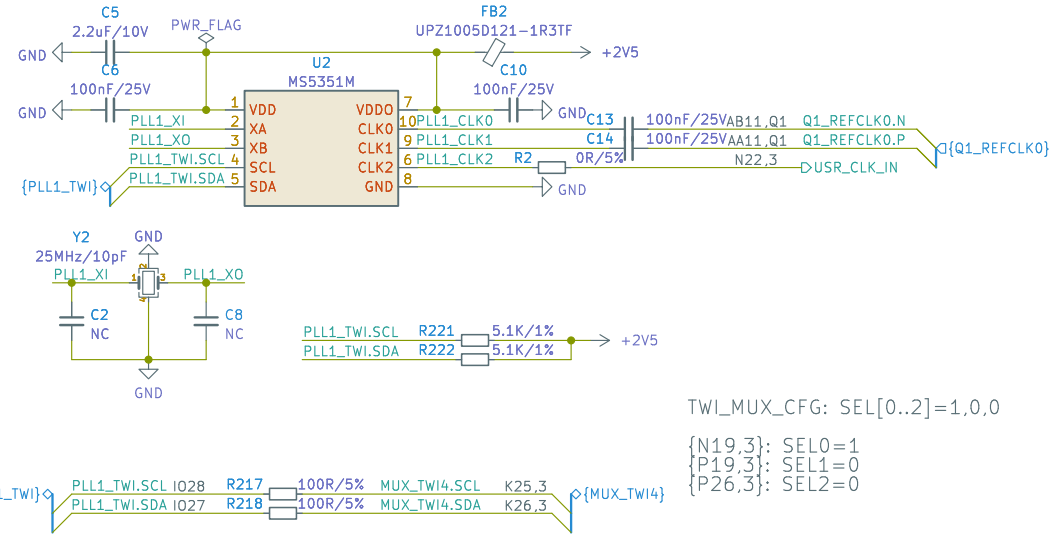
PLL0



TWI_MUX_CFG: SEL[0..2]=0,1,1

{N19,3}: SEL0=0
{P19,3}: SEL1=1
{P26,3}: SEL2=1

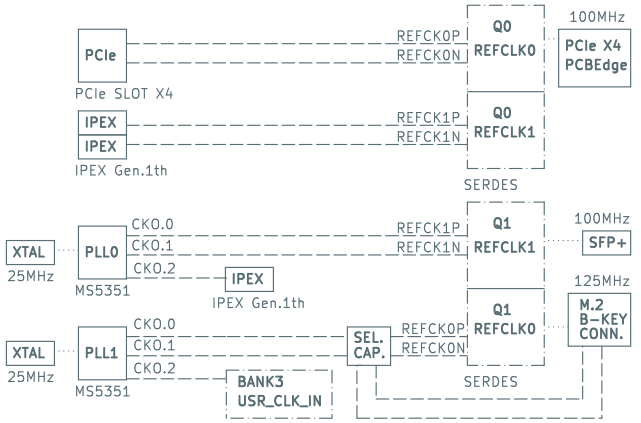
PLL1



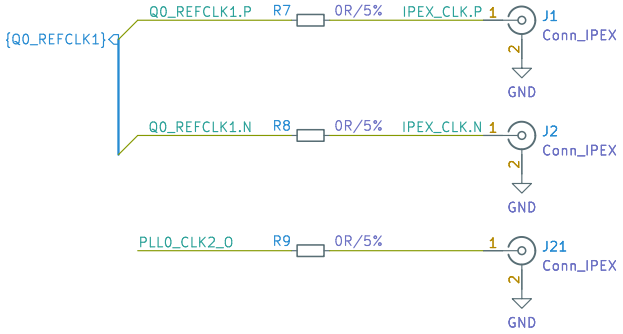
TWI_MUX_CFG: SEL[0..2]=1,0,0

{N19,3}: SEL0=1
{P19,3}: SEL1=0
{P26,3}: SEL2=0

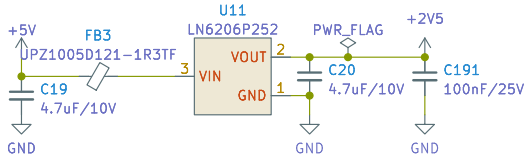
COLCK TREE



IPEX RF CONN.



EXT. PLL POWER

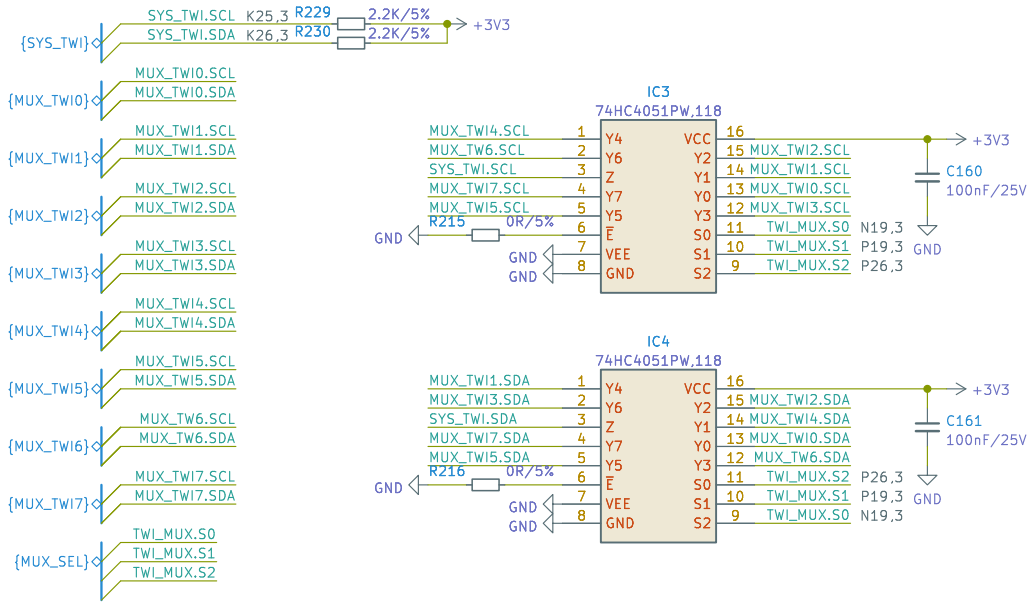


Update notes: Fixed the issues of clock tree.

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Sheet: /FPGA_CLOCK/		
File: CLOCK.kicad_sch		
Title: TANG_MEGA-138K-Pro_Dock-4071f_Schematics		
Size: A4	Date: 2024-10-10	**Rev: 1.2**
KiCad E.D.A. 8.0.5		Id: 3/19

8-WAY TWI MUX



NO.	CONFIG	SEL0	SEL1	SEL2
0	SFP0	0	0	0
1	SFP1	0	0	1
2	EEPROM	0	1	0
3	DVI_A_DDC MS5331-0	0	1	1
4	DVI_B_DDC MS5331-1	1	0	0
5	DVP_CAM	1	0	1
6	MIPI_RX0	1	1	0
7	MIPI_RX1	1	1	1

Update notes: Rename the DVI_TX_DDC to DVI_A_DDC, DVI_RX_DDC to DVI_B_DDC.

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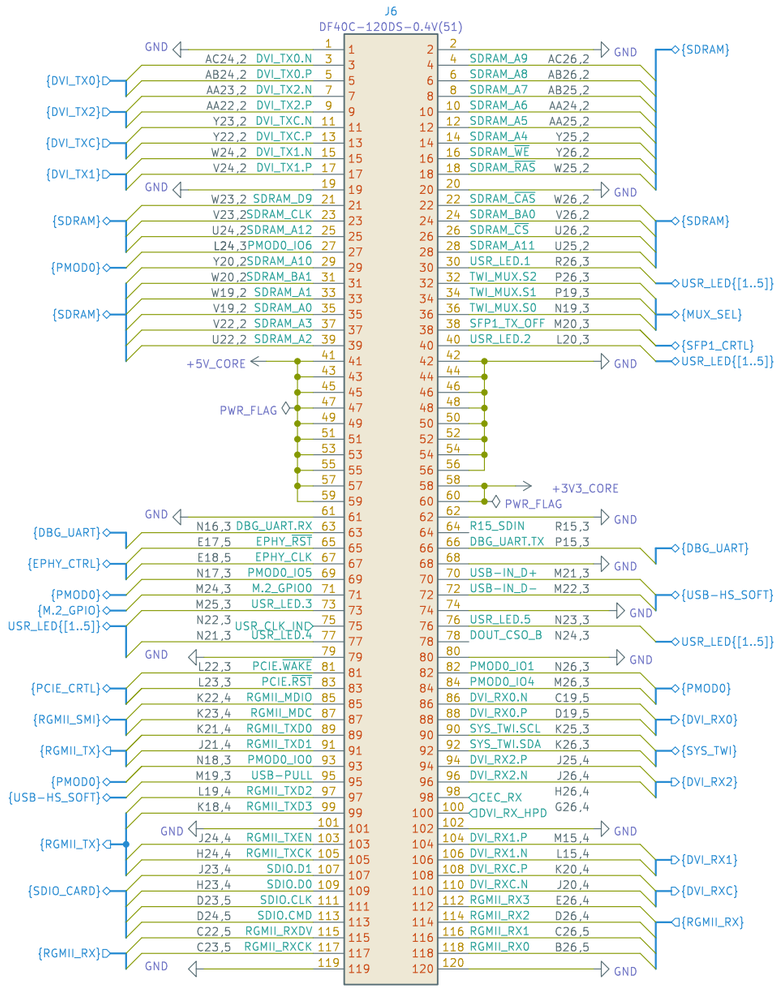
Sheet: /TWI_MUX/
File: TWI_MUX.kicad_sch

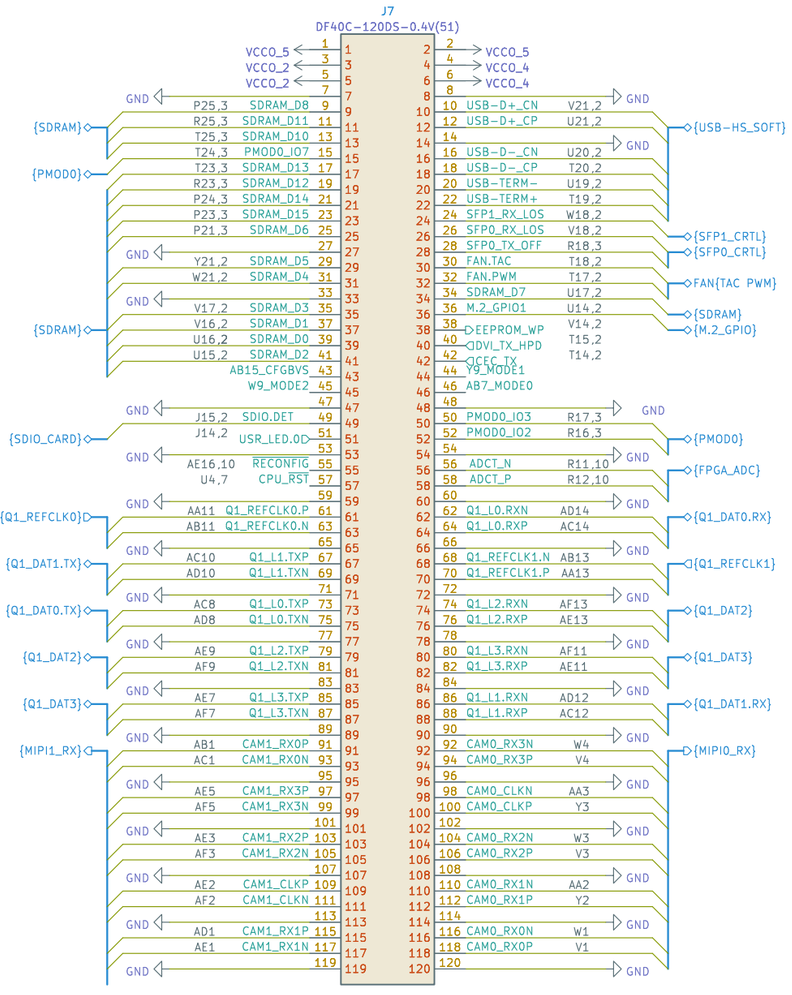
Title: TANG_MEGA-138K-Pro_Dock-4071f_Schematics

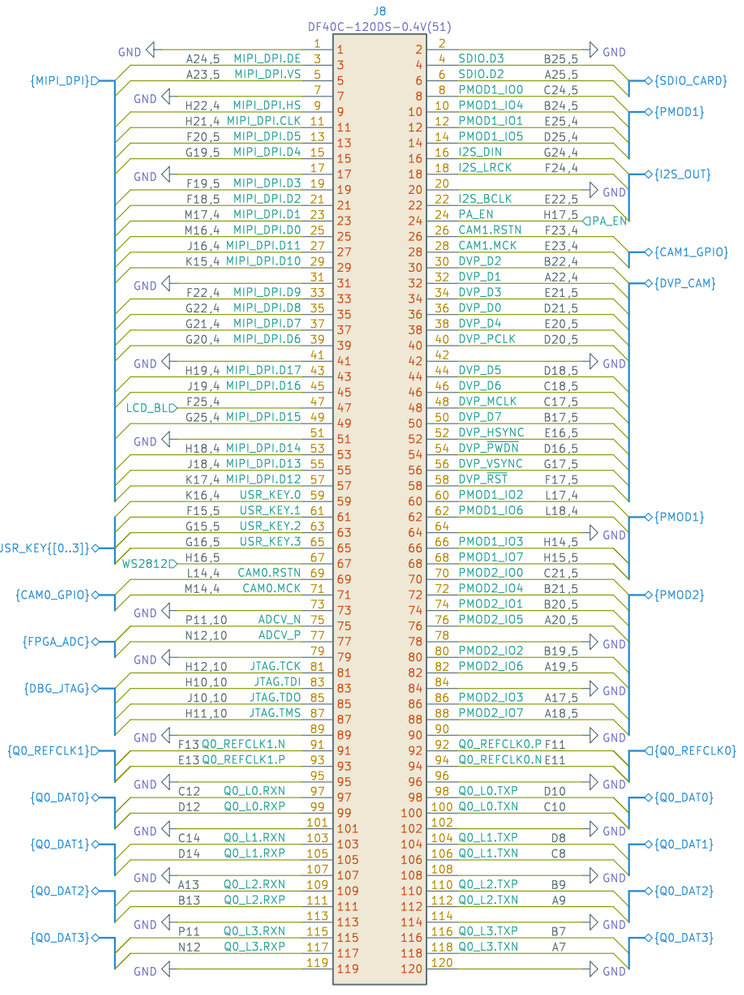
Size: A4	Date: 2024-10-10	**Rev: 1.2**
KiCad E.D.A. 8.0.5		Id: 4/19

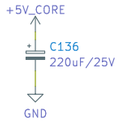
FPGA CORE MODULE

Note: that the pinout (numbers) of these conn. here is mirrored compared to the SOM.

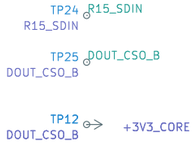






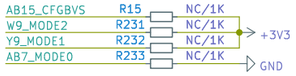


TEST POINT



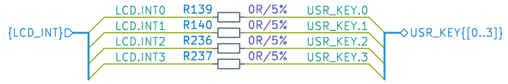


MODE SET



NOTE: MODE[0..2] is set to 001 by defualt on CM Board
Set CFGBVS to 1 to enable maximum driving of the pin during FPGA configuration

CONFIG	MODE0	MODE1	MODE2
JTAG	X	X	X
MSPI	0	0	1
SSPI	0	1	0
MSERIAL	0	0	0
SSERIAL	1	1	1
MCPU	1	0	0
SCPU	1	1	0

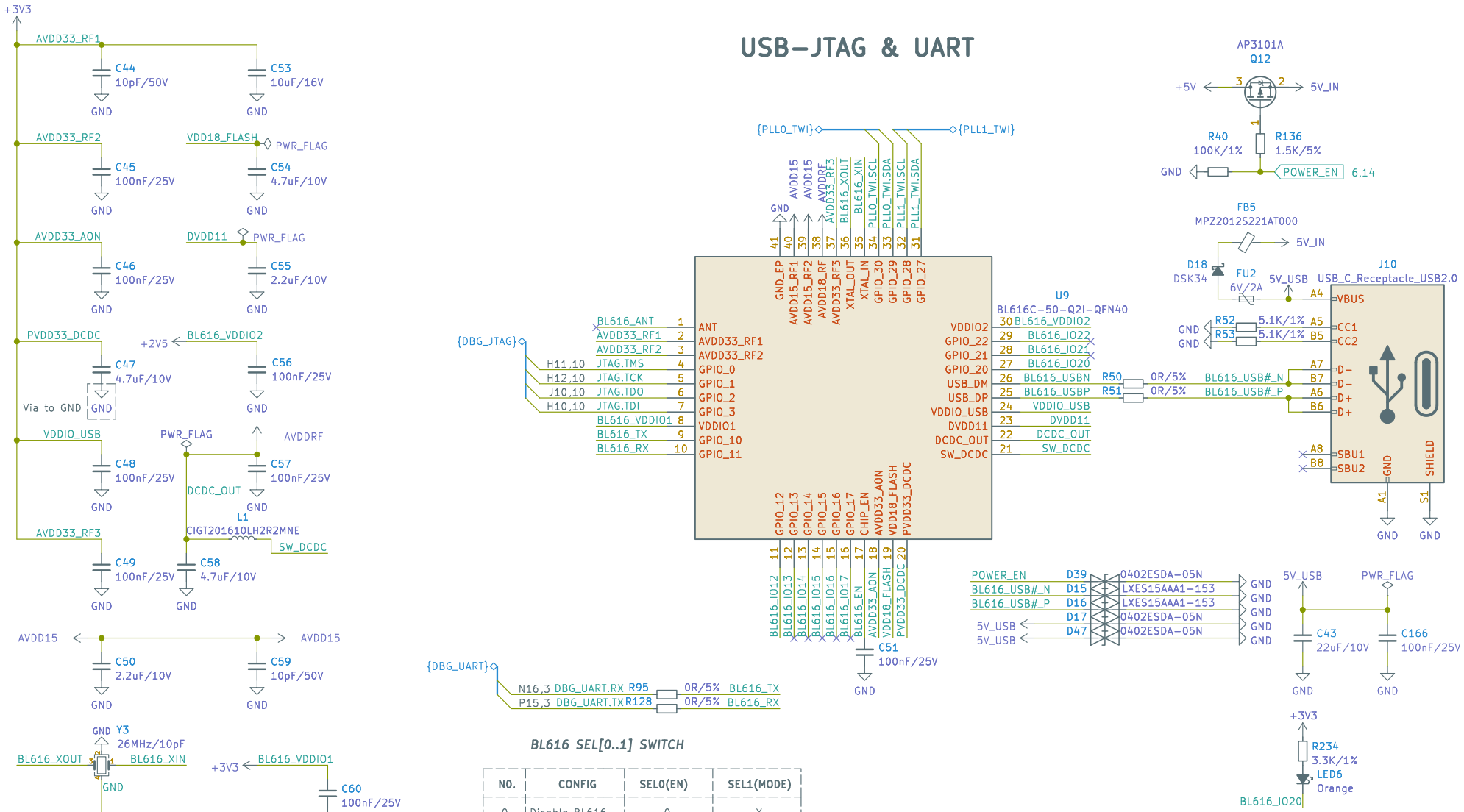


Update notes: Rename the DVI_TX_DDC DVI_A_DDC, DVI_RX_DDC to DVI_B_DDC.

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Sheet: /FPGA_BTB/		
File: CONN_BTB.kicad_sch		
Title: TANG_MEGA-138K-Pro_Dock-4071f_Schematics		
Size: A3	Date: 2024-10-10	**Rev: 1.2**
KiCad E.D.A. 8.0.5		Id: 5/19

USB-JTAG & UART



NO.	CONFIG	SEL0(EN)	SEL1(MODE)
0	Disable BL616	0	X
1	BL616 as Device	1	1
2	BL616 as Host	1	0

Update notes: Rename the DVI_TX_DDC DVI_A_DDC, DVI_RX_DDC to DVI_B_DDC.

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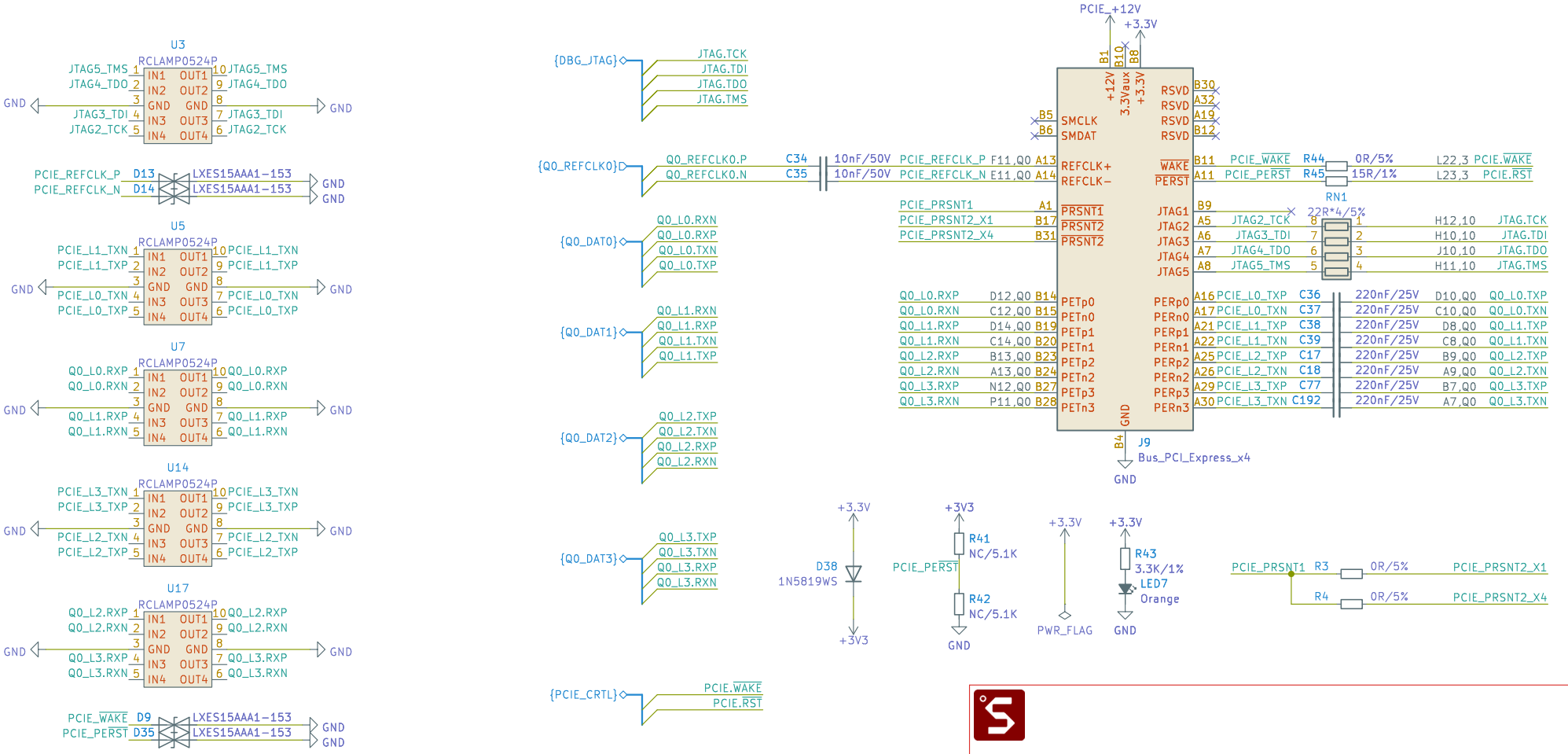
Sheet: /USB_JTAG/

File: USB_JTAG.kicad_sch

Title: TANG_MEGA-138K-Pro_Dock-4071f_Schematics

Size: A4	Date: 2024-10-10	**Rev: 1.2**
KiCad E.D.A. 8.0.5		Id: 6/19

PCI-Express X4



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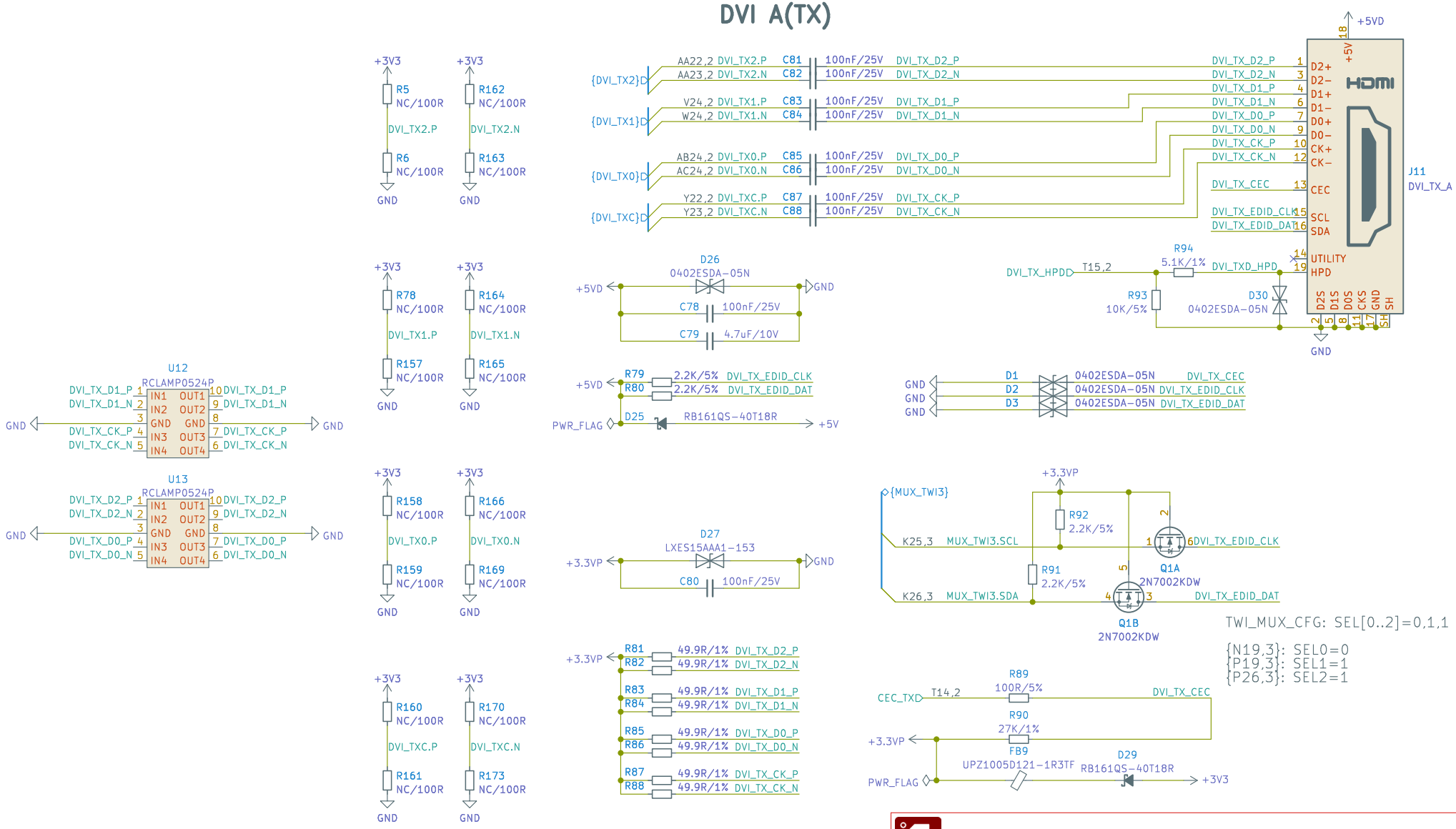
Sheet: /FPGA_PCIE/

File: PCIE_sch.kicad_sch

Title: TANG_MEGA-138K-Pro_Dock-4071f_Schematics

Size: A4	Date: 2024-10-10	**Rev: 1.2**
KiCad E.D.A. 8.0.5		Id: 7/19

DVI A(TX)



Update notes: Rename the DVI_TX to DVI_A(TX).

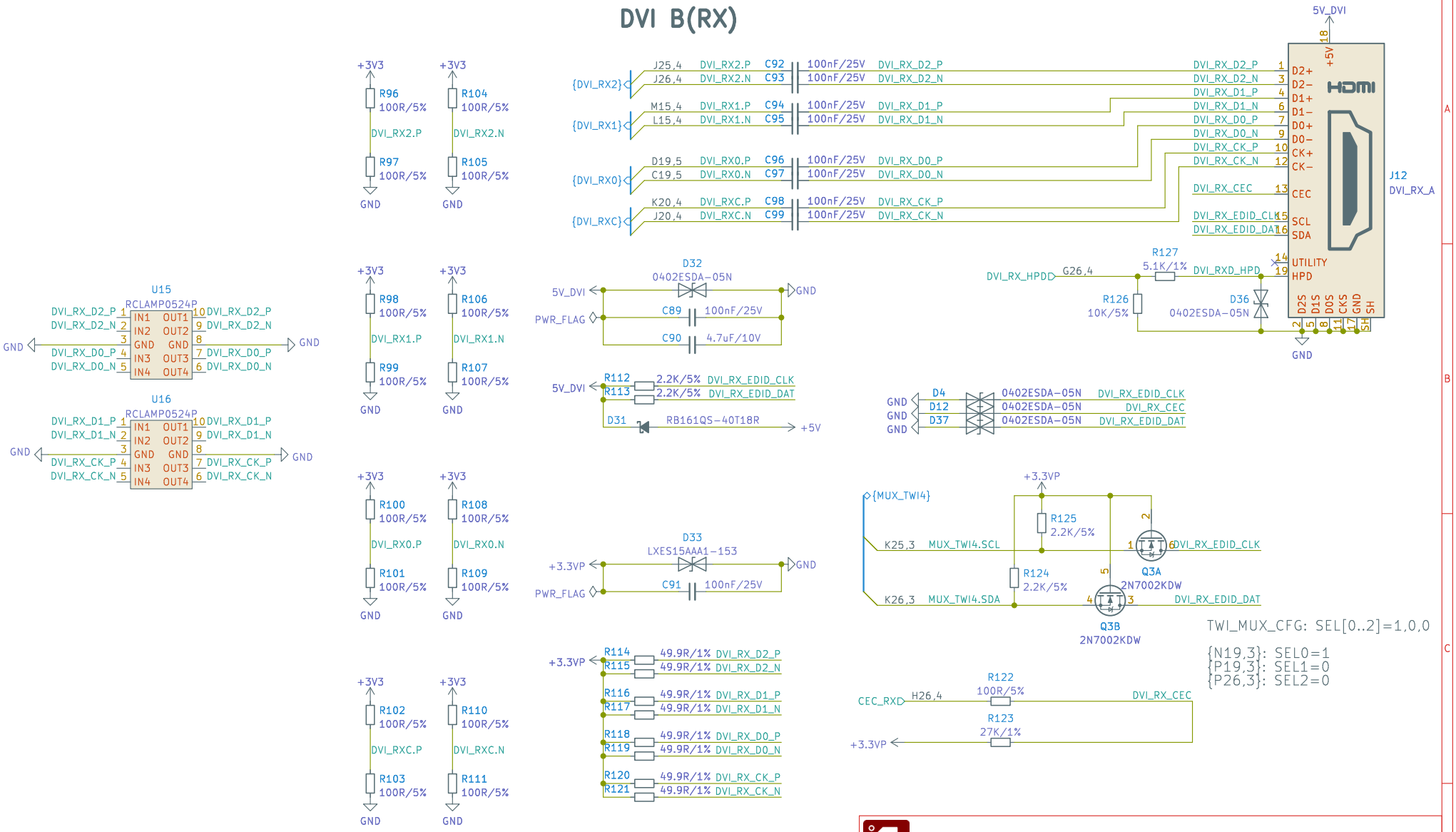
Shenzhen Sipeed Technology Co,Ltd.

Sheet: /FPGA_TMDS_A/
File: DVI_TX.kicad_sch

Title: TANG_MEGA-138K-Pro_Dock-4071f_Schematics

Size: A4	Date: 2024-10-10	**Rev: 1.2**
KiCad E.D.A. 8.0.5		Id: 8/19

DVI B(RX)



Update notes: Rename the DVI_RX to DVI_B(RX)

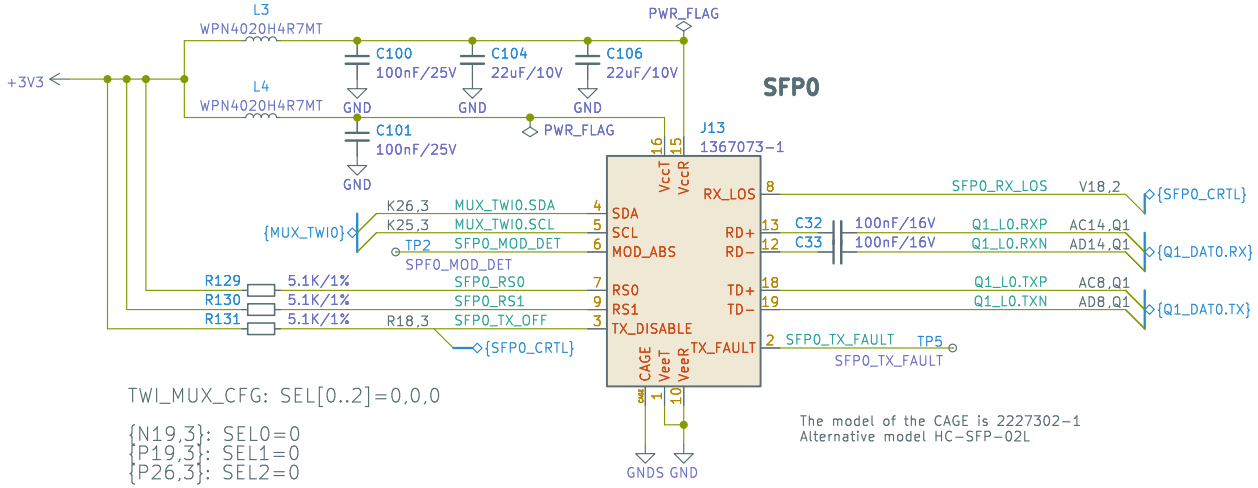
Shenzhen Sipeed Technology Co,Ltd.

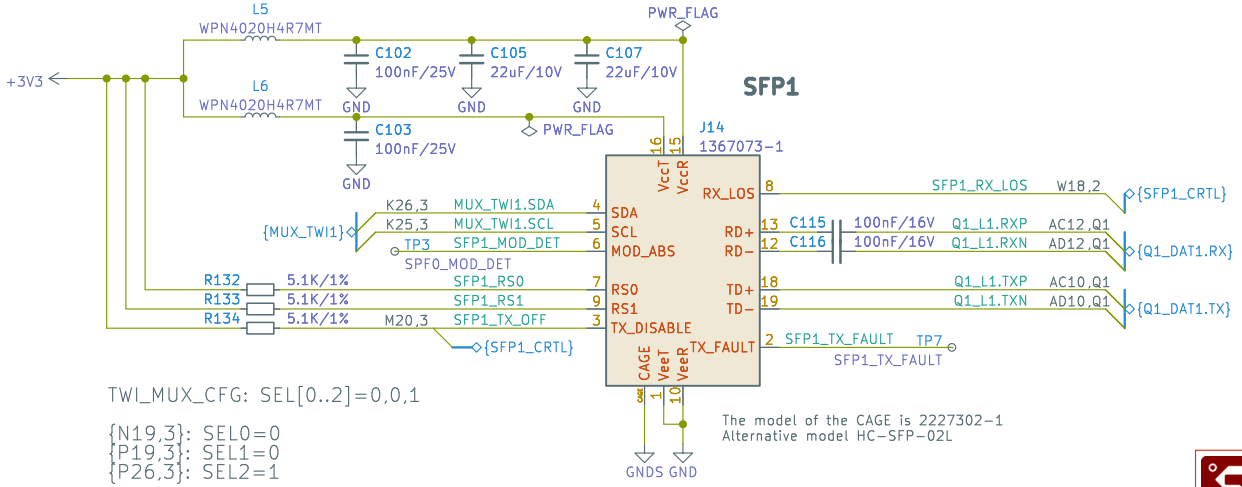
Sheet: /FPGA_TMDS_B/
File: DVI_RX.kicad_sch

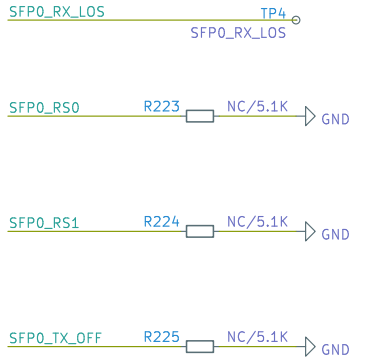
Title: TANG_MEGA-138K-Pro_Dock-4071f_Schematics

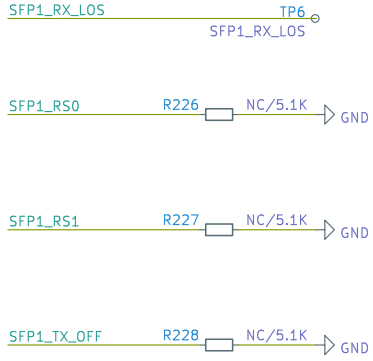
Size: A4	Date: 2024-10-10	**Rev: 1.2**
KiCad E.D.A. 8.0.5		Id: 9/19

SFP+









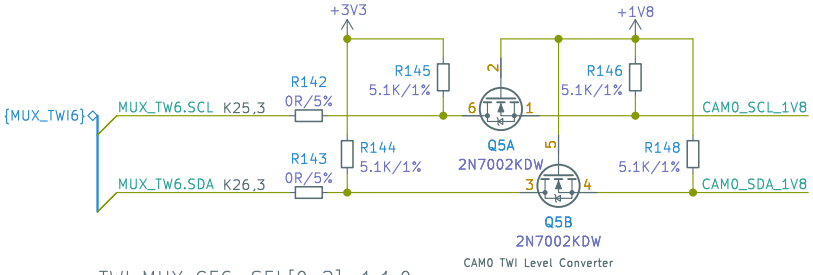
Shenzhen Sipeed Technology Co,Ltd.

Sheet: /FPGA_SFP+/
File: SFP-PLUS.kicad_sch

Title: TANG_MEGA-138K-Pro_Dock-4071f_Schematics

Size: A4	Date: 2024-10-10	**Rev: 1.2**
KiCad E.D.A. 8.0.5		Id: 10/19

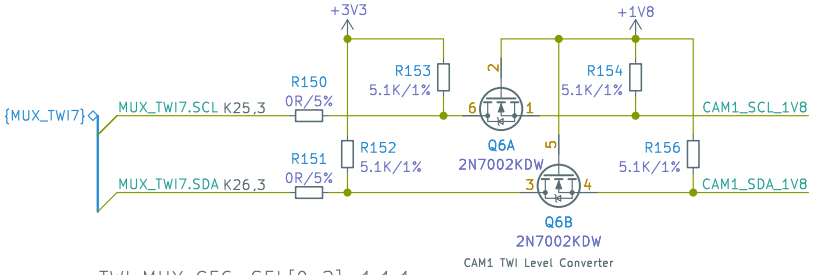
MIPI RX



CAM0 TWI Level Converter

TWI_MUX_CFG: SEL[0..2]=1,1,0

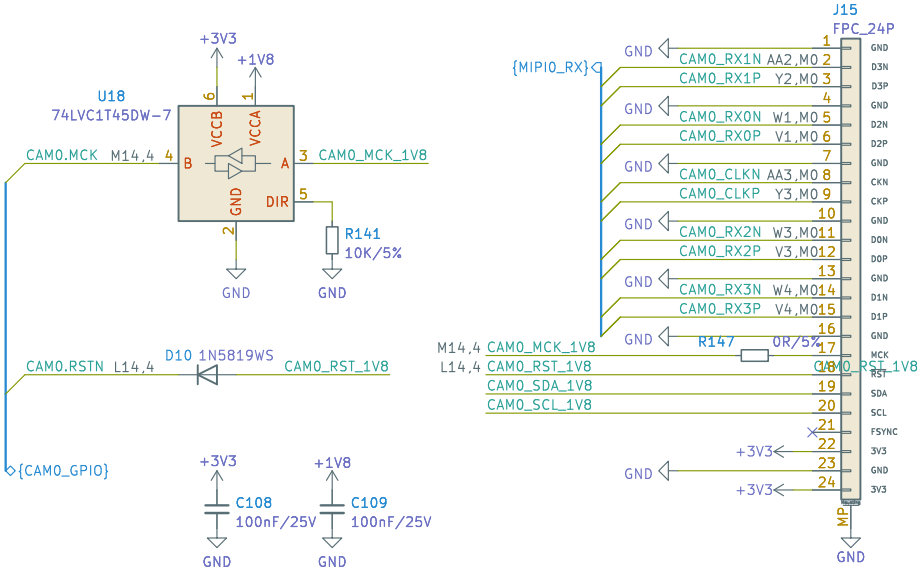
{N19,3}: SEL0=1
{P19,3}: SEL1=1
{P26,3}: SEL2=0

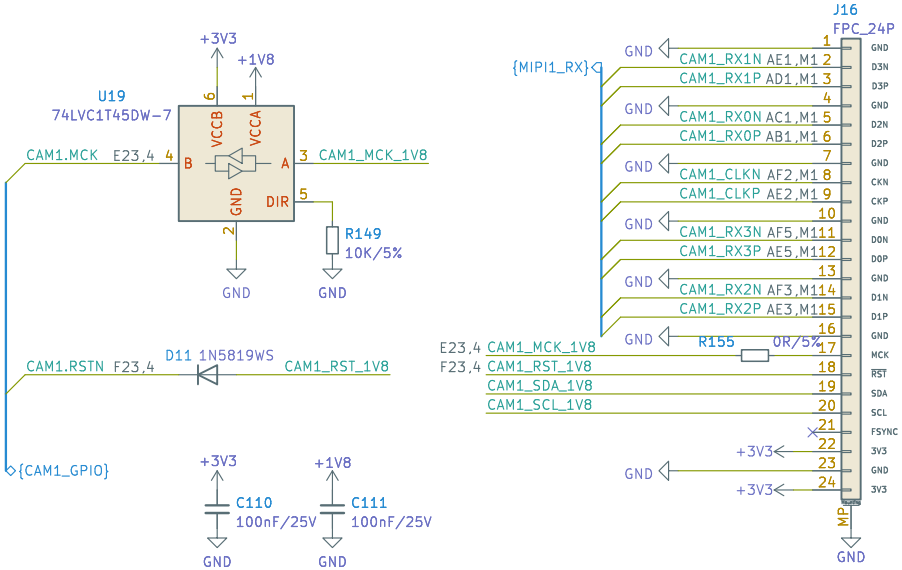


CAM1 TWI Level Converter

TWI_MUX_CFG: SEL[0..2]=1,1,1

{N19,3}: SEL0=1
{P19,3}: SEL1=1
{P26,3}: SEL2=1





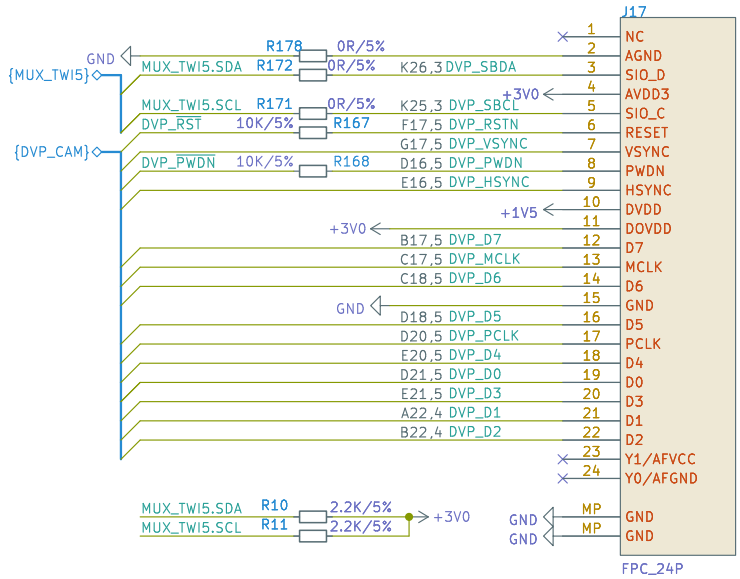
Shenzhen Sipeed Technology Co,Ltd.

Sheet: /FPGA_MIPI_RX/
File: MIPI_RX.kicad_sch

Title: TANG_MEGA-138K-Pro_Dock-4071f_Schematics

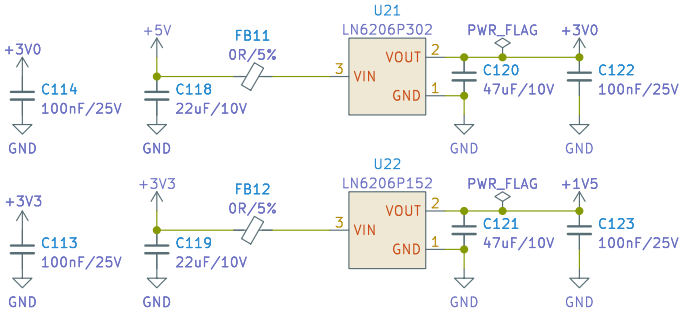
Size: A4	Date: 2024-10-10	**Rev: 1.2**
KiCad E.D.A. 8.0.5		Id: 11/19

DVP-CAMERA



TWI_MUX_CFG: SEL[0..2]=1,0,1

{N19,3}: SEL0=1
{P19,3}: SEL1=0
{P26,3}: SEL2=1



Shenzhen Sipeed Technology Co,Ltd.

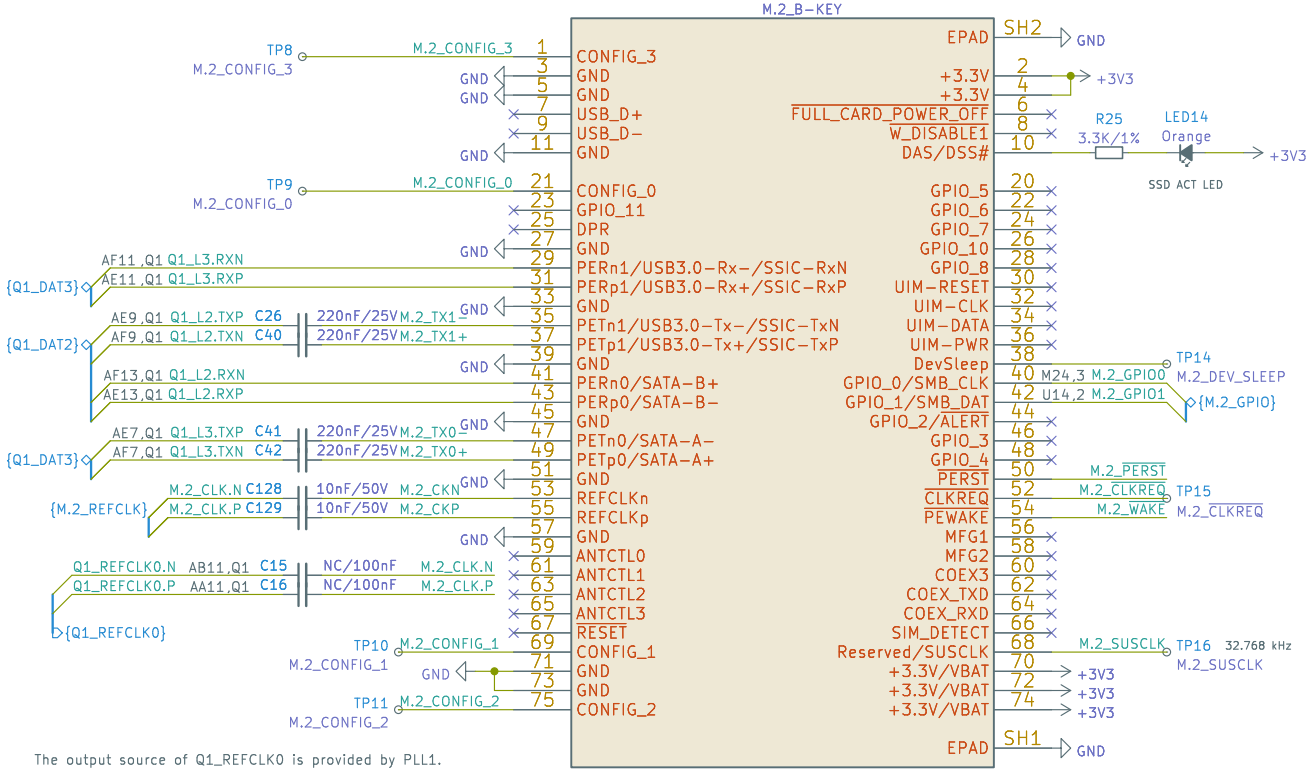
Sheet: /DVP_CAMERA/
File: DVP.kicad_sch

Title: TANG_MEGA-138K-Pro_Dock-4071f_Schematics

Size: A4	Date: 2024-10-10	**Rev: 1.2**
KiCad E.D.A. 8.0.5		Id: 12/19

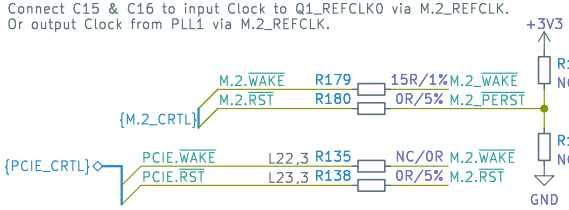
STORAGE

M.2 SOCKET KEY-B

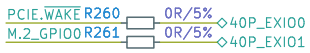


The output source of Q1_REFCLK0 is provided by PLL1.

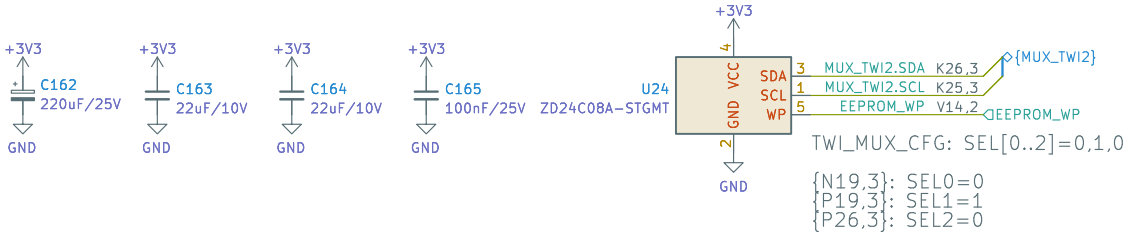
Connect C15 & C16 to input Clock to Q1_REFCLK0 via M.2_REFCLK.
Or output Clock from PLL1 via M.2_REFCLK.



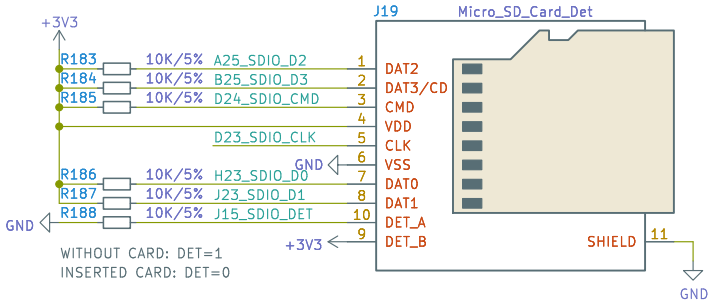


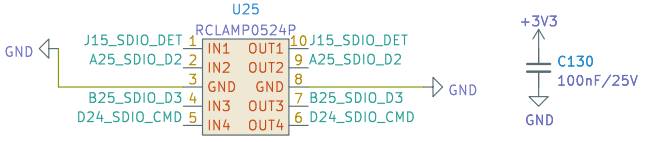


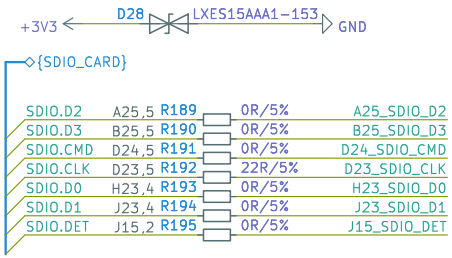
EEPROM



MICRO SD CARD



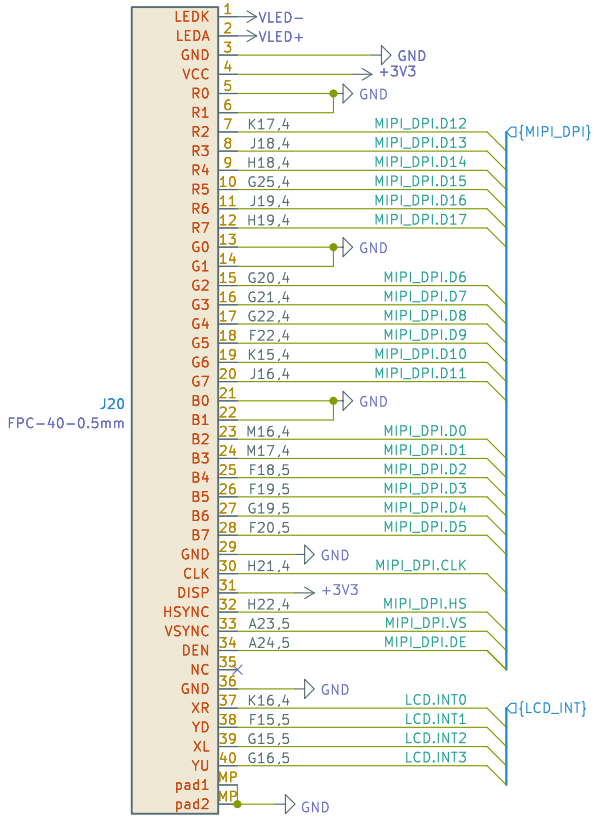




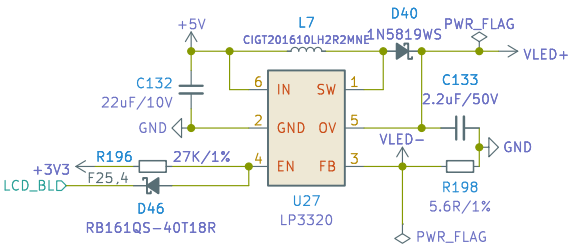
Shenzhen Sipeed Technology Co,Ltd.	
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Size: A4	Date: 2024-10-10
KiCad E.D.A. 8.0.5	Id: 13/19
	Rev: 1.2

FPGA-LCD

MIPI DPI CONN.



LCD BACKLIGHT

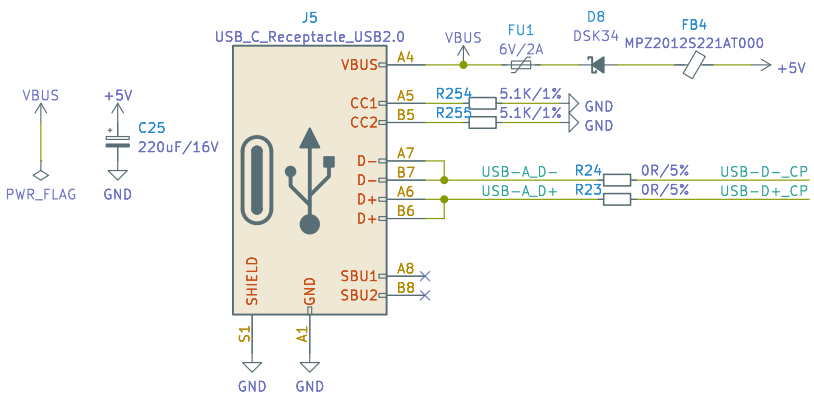


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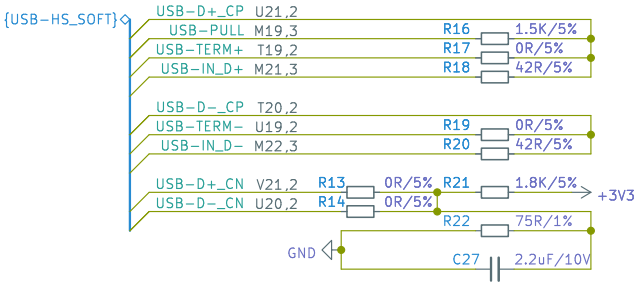
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Size: A4	Date: 2024-10-10	**Rev: 1.2**
KiCad E.D.A. 8.0.5		Id: 14/19

PERIPHERALS

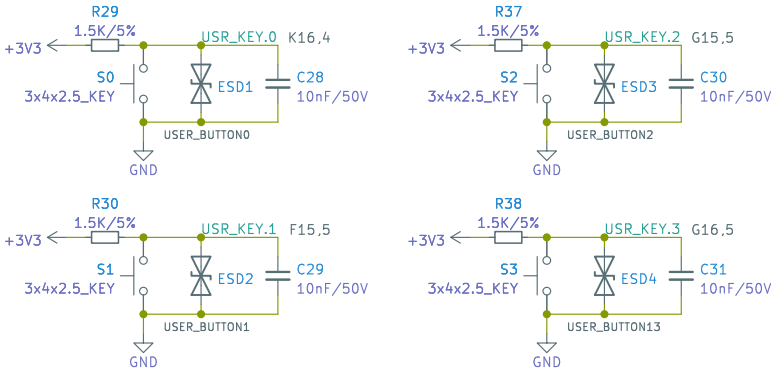
USB2.0-DEV.

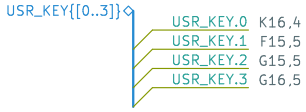




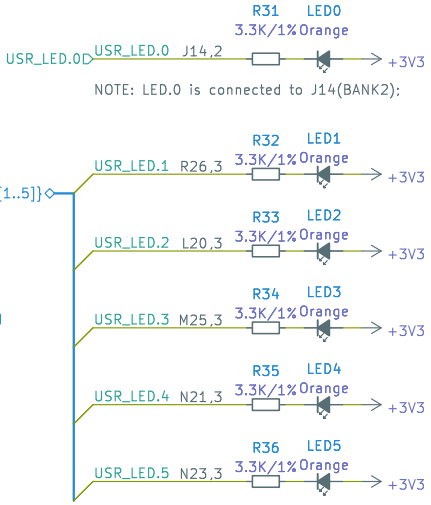


BUTTON

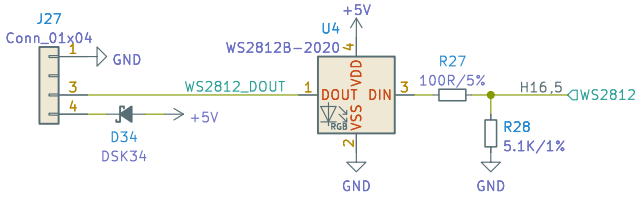




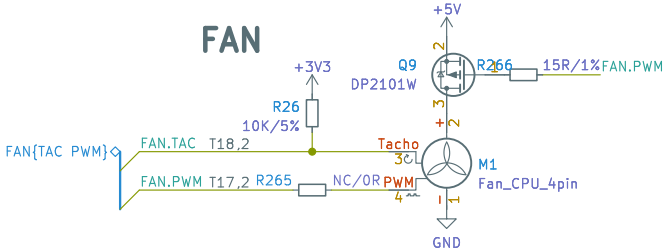
LED



WS2812



FAN



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Sheet: /PERIPHERALS/

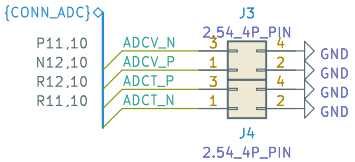
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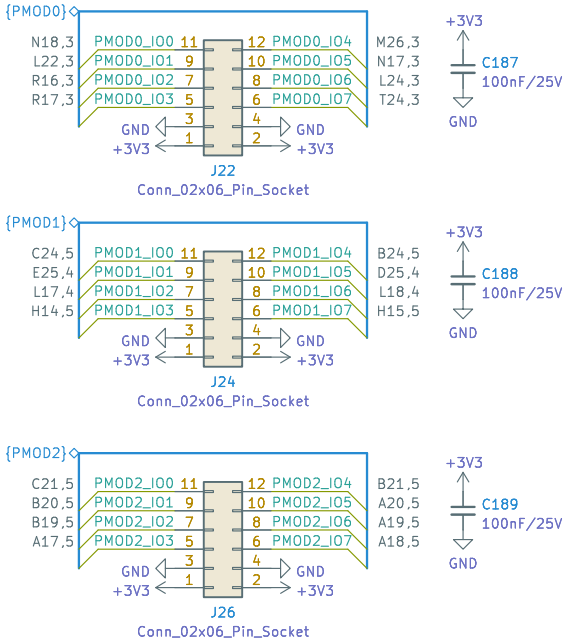
Size: A4	Date: 2024-10-10	**Rev: 1.2**
KiCad E.D.A. 8.0.5		Id: 15/19

EX CONN.

ADC CONN.



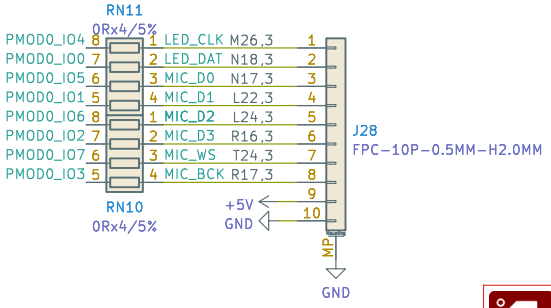
PMOD



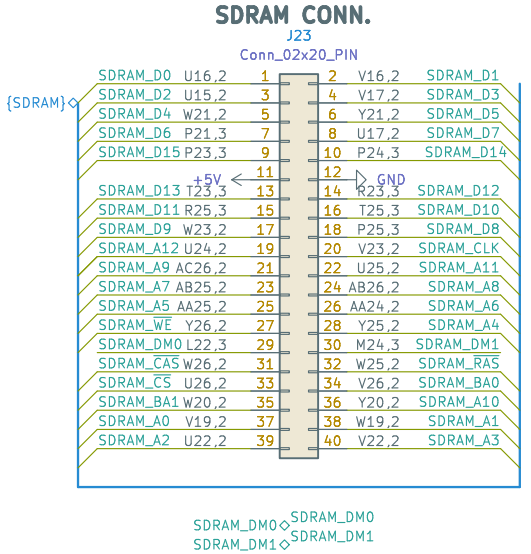
SSPI

PIN	FUCTION	PMOD0
N18	SSPI_SO	IO0
R16	SSPI_CLK	IO2
R17	SSPI_WP	IO3
N17	SSPI_SI	IO5
L24	SSPI_CS	IO6
T24	SSPI_HOLD	IO7

MIC-ARRAY CONN.



SDRAM CONN.



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Sheet: /FPGA_EXT_CONN./

File: EX_CONN..kicad_sch

Title: TANG_MEGA-138K-Pro_Dock-4071f_Schematics

Size: A4	Date: 2024-10-10	**Rev: 1.2**
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RGMII—ETHERNET

U10
RTL8211FI—CG

PHY0_MDI0+ 1 MDIP0
PHY0_MDI0— 2 MDIN0
PHY0_MDI1+ 4 MDIP1
PHY0_MDI1— 5 MDIN1
PHY0_MDI2+ 6 MDIP2
PHY0_MDI2— 7 MDIN2
PHY0_MDI3+ 9 MDIP3
PHY0_MDI3— 10 MDIN3

GND R54 0R/5% PHY0_CLK 36 XTAL_IN
37 XTAL_OUT/EXT_CLK
TP1 C0P8 35 CLKOUT

PHY0_TXD0 18 TXD0
PHY0_TXD1 17 TXD1
PHY0_TXD2 16 TXD2
PHY0_TXD3 15 TXD3

PHY0_TXEN 19 TXCTL
PHY0_TXCLK 20 TXCLK

PHY0_RXD0/RXDLY 25 RXD0/RXDLY
PHY0_RXD1/TXDLY 24 RXD1/TXDLY
PHY0_RXD2/PLLOFF 23 RXD2/PLLOFF
PHY0_RXD3/AD0 22 RXD3/PHYAD0

PHY0_RXCLK/AD1 27 RXC/PHYAD1
PHY0_RXDV/AD2 26 RXCTL/PHY_AD2

13 MDC ← PHY0_MDCK
14 MDIO ← PHY0_MDIO
R70 1.5K/5% PHY_VIO
12 PHYRSTB → PHY0_RSTN
31 INTB/PMEB
R61 NC/5.1K GND
C64 10nF/50V GND

32 LED0/CFG_EXT → PHY0_LED0/CFG_EXT
33 LED1/CFG_LDO0 → PHY0_LED1/CFG_LDO0
34 LED2/CFG_LDO1 → PHY0_LED2/CFG_LDO1
39 RSET
R62 2.49K/1%
GND

30 REG_OUT/LDO_OUT
PHY0_REG_SW
L2 CIGT201610LH2R2MNE
PHY0_REG_1V0
C65 4.7uF/10V GND
C67 100nF/25V GND

29 DVDD33 → PHY0_DVDD_3V3
21 DVDD10 → PHY0_DVDD_1V0
28 DVDD_RG → PHY_VIO

3 AVDD10_1
8 AVDD10_2
38 AVDD10_3
→ PHY0_AVDD_1V0

11 AVDD33_1
40 AVDD33_2
→ PHY0_AVDD_3V3

41 GND → GND

RN6 0Rx4/5%
RN7 0Rx4/5% PWR_FLAG
GND GNDS

+3V3 FB7 UPZ1005D121—1R3TF PWR_FLAG PHY0_AVDD_3V3
C69 100nF/25V GND
C70 100nF/25V

+3V3 FB8 UPZ1005D121—1R3TF PWR_FLAG PHY0_DVDD_3V3
C73 4.7uF/10V GND
C75 100nF/25V

PHY0_REG_1V0 PHY0_DVDD_1V0
R74 0R/5% PWR_FLAG
PWR_FLAG C68 100nF/25V GND

PHY0_REG_1V0
R75 0R/5% PWR_FLAG PHY0_AVDD_1V0
PWR_FLAG C72 100nF/25V C74 100nF/25V C76 100nF/25V GND

C71 10nF/50V GND

RJ1 HY911130A
1 SH 16 GNDS
2 SH 15
Yellow 14 PHY0_LED2 GNDS
Green 12 PHY0_LED1 GNDS

RN8 2.2Rx4/1%
RN9 2.2Rx4/1%
PHY0_MDI0+ 8 1 R_MDI0+ D19 UDD32C03L01 R_MDI0—
PHY0_MDI0— 7 2
PHY0_MDI1+ 6 3 R_MDI1+ D20 UDD32C03L01 R_MDI1—
PHY0_MDI1— 5 4
PHY0_MDI2+ 8 1 R_MDI2+ D21 UDD32C03L01 R_MDI2—
PHY0_MDI2— 7 2
PHY0_MDI3+ 6 3 R_MDI3+ D22 UDD32C03L01 R_MDI3—
PHY0_MDI3— 5 4
GNDS

{RGMII_TX}
Resistors should be placed close to SOC
RN2 0Rx4/5%
RGMII_TXD0 K21,4 8 1 PHY0_TXD0
RGMII_TXD1 J21,4 7 2 PHY0_TXD1
RGMII_TXD2 L19,4 6 3 PHY0_TXD2
RGMII_TXD3 K18,4 5 4 PHY0_TXD3

RN3 0Rx4/5%
RGMII_TXCK H24,4 8 1 PHY0_TXCLK
RGMII_TXEN J24,4 7 2 PHY0_TXEN
RGMII_MDIO K22,4 6 3 PHY0_MDIO
RGMII_MDC K23,4 5 4 PHY0_MDCK
{RGMII_SMI}

{RGMII_RX}
Resistors should be placed close to PHY
RN4 0Rx4/5%
RGMII_RX0 B26,5 8 1 PHY0_RXD0/RXDLY
RGMII_RX1 C26,5 7 2 PHY0_RXD1/TXDLY
RGMII_RX2 D26,4 6 3 PHY0_RXD2/PLLOFF
RGMII_RX3 E26,4 5 4 PHY0_RXD3/AD0

RN5 0Rx4/5%
RGMII_RXDV C22,5 8 1 PHY0_RXDV/AD2
RGMII_RXCK C23,5 7 2 PHY0_RXCLK/AD1
EPHY_CLK E18,5 6 3 PHY0_CLK
EPHY_RST E17,5 5 4 PHY0_RSTN
{EPHY_CTRL}

+3V3 FB6 UPZ1005D121—1R3TF PWR_FLAG PHY_VIO
C61 100nF/25V GND
C62 4.7uF/10V GND
C63 100nF/25V GND
C66 100nF/25V GND

PHY0 Config
PHY0 Address = 3'b001
PHY_VIO
R55 NC/5.1K PHY0_RXDV/AD2 R63 5.1K/1%
R56 NC/5.1K PHY0_RXCLK/AD1 R64 5.1K/1%
R57 5.1K/1% PHY0_RXD3/AD0 R65 NC/5.1K
PHY0 Address = 3'b001
R58 NC/5.1K PHY0_RXD2/PLLOFF R66 5.1K/1%
Pull Down, PPL coontinue toggling@ALDPS Mode
R59 5.1K/1% PHY0_RXD1/TXDLY R67 NC/5.1K
R60 5.1K/1% PHY0_RXD0/RXDLY R68 NC/5.1K
Pull Up to add 2ns delay to TXC/RXC for TXD/RXD latching
GND

R76 510R/5% PHY0_LED2 PHY0_LED2/CFG_LDO1 LED_1000M
D23 LXES15AAA1—153
PHY0_LED1 D24 LXES15AAA1—153 GNDS
PHY0_LED1/CFG_LDO0 LED_100M
R77 510R/5%

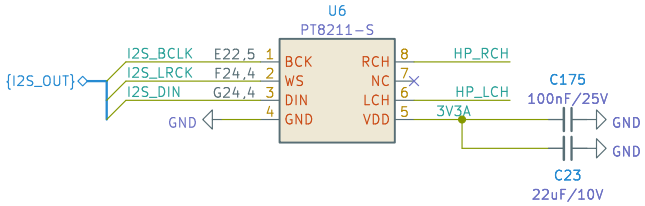
RGMII IO Power	CFG_EXT	CFG_LDO[1:0]
External 3.3V	1'b1	2'b00
External 1.8V	1'b1	2'b10
Internal 1.8V	1'b0	2'b10

PHY_VIO R71 5.1K/1% PHY0_LED0/CFG_EXT
Pull up to use external power source for IO
PHY0_LED2/CFG_LDO1 R72 5.1K/1%
PHY0_LED1/CFG_LDO0 R73 5.1K/1% GNDS
CFG_LDO[1:1]=0x11, select 3.3V external power

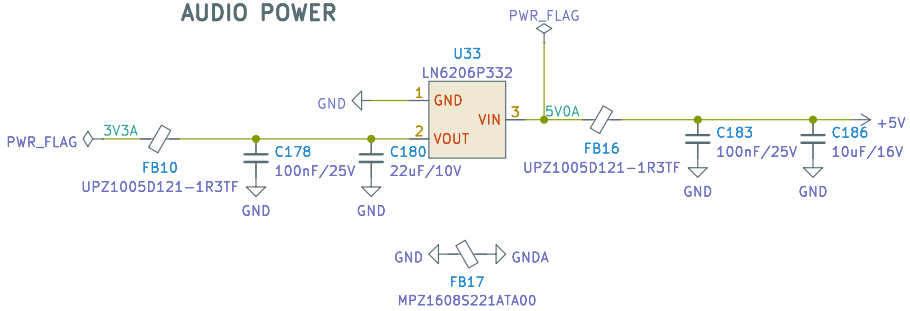
Shenzhen Sipeed Technology Co,Ltd.
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Size: A4 | Date: 2024—10—10 | Rev: 1.2
KiCad E.D.A. 8.0.5 | Id: 17/19

FPGA-Audio

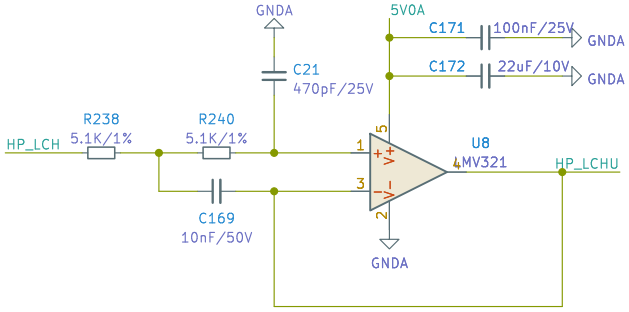
STEREO DAC

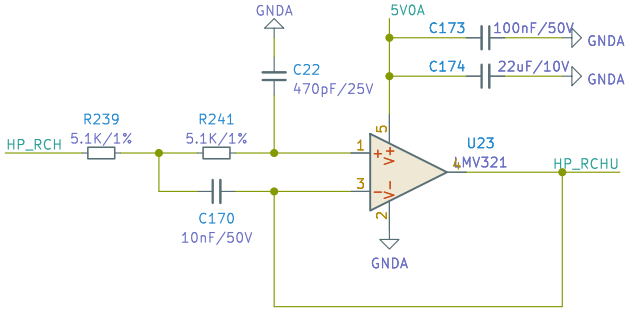


AUDIO POWER

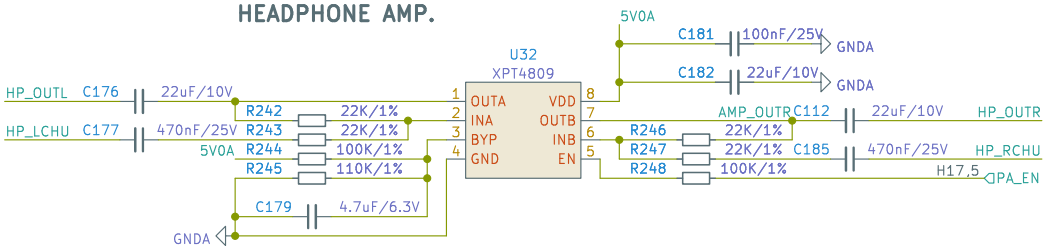


LOW-PASS FILTER

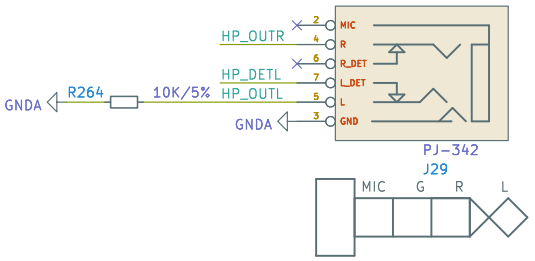




HEADPHONE AMP.

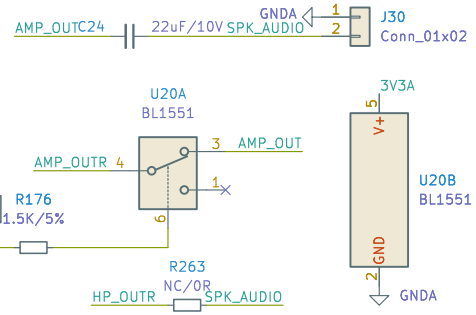


HEADPHONE JACK 3.5MM CTIA



When the jack is unpluged, L/R_DET is connected to L/R.

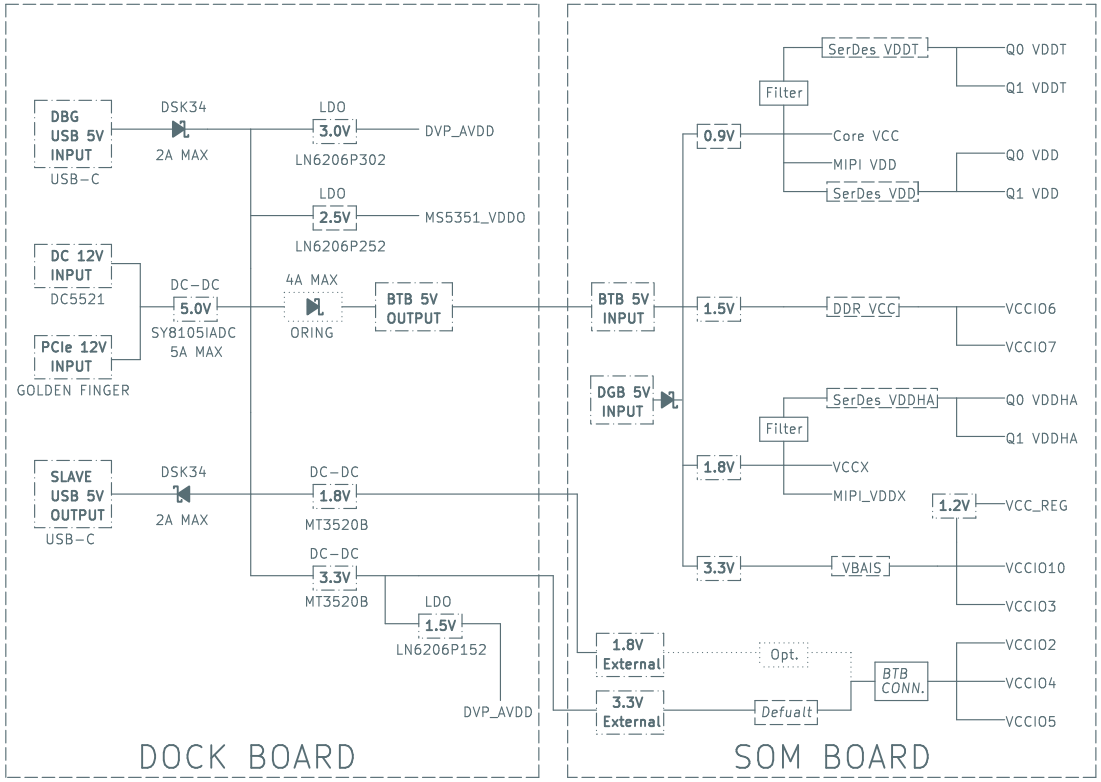
SPEAKER CONN. R-CH



Update notes: Added DC bias to the AUDIO_OUTR network to work with analog switches.

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Title: TANG_MEGA-138K-Pro_Dock-4071f_Schematics		
Size: A4	Date: 2024-10-10	**Rev: 1.2**
KiCad E.D.A. 8.0.5		Id: 18/19



Update notes: Rename the DVI_TX_DDC DVI_A_DDC, DVI_RX_DDC to DVI_B_DDC.

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Sheet: /POWER_TREE/
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Title: TANG_MEGA-138K-Pro_Dock-4071f_Schematics

Size: A4	Date: 2024-10-10	**Rev: 1.2**
KiCad E.D.A. 8.0.5		Id: 19/19